

2025 Tests

Semester Test 1

Question 1

In each case, select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1. A proprietary network protocol is a protocol that:
 - A. Has been standardised by some national standards body.
 - B. Is owned by a specific vendor.
 - C. Is developed as an open-source protocol.
 - D. Is now so old that using it has been deprecated.
 - E. Describes the properties of other network protocols.

2. Which historical event had a profound impact on many design decisions for the Internet (or ARPAnet then)?
 - A. Cold War
 - B. Suez Canal Crisis
 - C. Rise of the Global South
 - D. Bay of Pigs Invasion
 - E. Lockerbie Bombing

3. When writing an application that will serve as a server in a client-server architecture, the application would (in its role as server) open a socket that would:
 - A. Be specified as an address and port number
 - B. Initiate connections
 - C. Wait for connections
 - D. More than one of the above
 - E. All of the above

4. The phrase *protocol stack* is used in a network context because:
 - A. Protocols are "stacked" in layers on top of one another.
 - B. A message is "pushed" down when transmitted and then "popped" at the receiving end, such that the most recently sent message is the first one that would be retrieved by the receiver.
 - C. Alternative protocols may exist "next to" one another within the layers of protocols. Examples include IPv4 and IPv6.
 - D. Many protocols on the higher layers (such as layer 4) tend to converge into one protocol (or very few protocols) on layer 4. From layer 4 it "spreads out" to many lower layer protocols (especially on layers 2 and 1).
 - E. The notion of a protocol stack is not used in networking.

5. ? The OSI layer that, amongst others, attempts to emulate database transactions where messages can be “rolled back” if the final message in a sequence is not sent, is layer:
- A. 2
 - B. 3
 - C. 4
 - D. 5
 - E. 6
6. Which ISO OSI layer is responsible for regulating which node is allowed to transmit next in cases where two computers are directly connected?
- A. 5
 - B. 4
 - C. 3
 - D. 2
 - E. It may be 2 or 5 depending on factors not mentioned in the question.
7. Protocols such as SMTP, POP3, and FTP are deemed to be application layer protocols. As such, they obviously include ISO OSI layer 7 functionality. In addition, they often include functionality associated with the following ISO OSI layer(s):
- A. Only layer 6
 - B. Layers 6 and 5
 - C. Layers 5 and 4
 - D. Layers 4 and 3
 - E. Only layer 3
8. The **nom.za** second-level domain is intended to allow ... to register subdomains within it.
- A. Nomads
 - B. Individuals
 - C. Parties who handle nominations for potential office bearers
 - D. Those who provide authoritative definitions of nomenclature
 - E. None, because no such second-level domain exists
9. The .io TLD is a(n):
- A. sTLD
 - B. ccTLD
 - C. gTLD
 - D. Reserved TLD
 - E. None of the above

10. A file on a name server that maps names to IP addresses is known as a ... file.
- A. Name
 - B. Hosts
 - C. Mapping
 - D. Resolution
 - E. None of the above
11. The capetown. sTLD has to adhere to policies prescribed by:
- A. ICANN
 - B. ZADNA
 - C. ISO
 - D. More than one of the above
 - E. All of the above
12. The primary metric used by interior gateway protocols is:
- A. Cost
 - B. Hop count
 - C. Distance
 - D. Static routing tables
 - E. Speed
 - F. Noise
13. Consider a scenario in RIP where a router r sends its routing table to a router s. An entry in r's table indicates that it can reach a network n at a cost of 5. Before receiving the message, the routing table at s indicated that it could reach n at a cost of 8. What will the cost from s to n be according to the routing table at s after s processes the message from r?
- A. 5
 - B. 6
 - C. 7
 - D. 8
 - E. More information about r is required to answer the question.
14. Which of the following protocols is/are examples of an **external gateway protocol (EGP)**?
- A. BGP
 - B. OSPF
 - C. RIP
 - D. More than one of the above
 - E. All of the above
15. Routers using **OSPF** broadcast their ... and use the received information to apply the ... algorithm.
- A. Routing tables; Dijkstra
 - B. Routing tables; Bellman-Ford
 - C. Routing tables; RIP
 - D. Version of the network topology; Dijkstra
 - E. Version of the network topology; Bellman-Ford

16. Which protocol is primarily intended to manipulate a mailbox on a server?
- A. IMAP4
 - B. POP3
 - C. SMTP
 - D. More than one of the above
 - E. All of the above
17. Which condition code will be returned by an SMTP server after successfully handling a HELO or EHLO message?
- A. 0
 - B. 150
 - C. 250
 - D. 350
 - E. 450
18. (i) How does a CGI program send output to a browser?
(ii) How does a CGI program receive input from a browser?
- A. (i) Writes to standard output; (ii) Reads from standard input
 - B. (i) Writes to the appropriate socket; (ii) Reads from standard input
 - C. (i) Writes to standard output; (ii) Reads from the appropriate socket
 - D. (i) Writes to the appropriate socket; (ii) Reads from the appropriate socket
 - E. (i) Writes to standard output; (ii) Reads an environment variable set by the server
19. To avoid the need for the ... channel to connect to the FTP client, FTP ... mode should be used.
- A. data, passive
 - B. data, indirect
 - C. data, reversed
 - D. control, passive
 - E. control, indirect
 - F. control, reversed
20. An agent that monitors and controls network activity in an SNMP-managed network node is a(n):
- A. Client
 - B. Server
 - C. P2P-node
 - D. More than one of the above
 - E. All of the above

Question 2 [5 marks]

Fill in the blanks in the following paragraph by providing the missing words or phrases on your answer sheet:

Sam finds that the domain **xx.co.za** was not renewed and is available for registration. Sam approaches **xneelo** to register this domain, which xneelo does. In this scenario, xneelo acts as the **[a]** of the domain when they add the domain to the **co.za** registry. The co.za registry is operated by **[b]**.

After registration, Sam performs a **whois** query for the domain on the registry; the addresses of three parties (who may be the same) are returned as a result. In sequence, the details of the **[c]**, as well as the information to contact the **[d]** and **[e]** "departments" associated with the domain. (In this particular case, the latter two responses may provide contact details at xneelo.)

However, in recent years, the returned information is often of little use, since it is typically **[f]**. As an aside, the whois service is usually associated with port **[g]**.

It may also be possible to retrieve whois information from a URL that uses domain name hacking and which is available at **https://[h]**.

Given that the xx.co.za is in South Africa, it, like other South African domain names, has to abide by the general Internet policies, as well as policies imposed by **[i]**. Recall that **za.** is a **[j]** TLD.

Question 3 [5 marks]

Compile a **zone file** for a name server for the domain **xx.co.za**.

- **Do not** include the SOA (Start of Authority) entry.
- Only provide the **other resource records (RRs)** that are required based on the specification below.
- Provide **one RR per line** on your answer sheet.
- **Omit** the "IN" entry (used to indicate the record is for the Internet).
- In most cases, a resource record will consist of a **triple**: **(name, type, value)** — but note the few exceptions to this format and handle them correctly.

Specification:

- xx.co.za exists as a **single computer** at address **41.203.18.59**.
- It hosts a web server accessible via https://xx.co.za.
- The same server is also accessible via https://www.xx.co.za, but the owner wants to avoid multiple RRs if the IP address changes.
- The **primary mail server**: ASPMX.L.GOOGLE.COM.
- The **secondary mail server**: ALT1.ASPMX.L.GOOGLE.COM.
- The **primary name server**: ns1.host-h.net.
- The **secondary name server**: ns2.host-h.net.
- The domain has a **subdomain** abc.xx.co.za, which uses the **same name servers** as the main domain.
- The **IPv6 address** of the computer is 2001::1234.
- The owner wants DNS to indicate that this machine is a **Raspberry Pi**.

The SOA record (which you are to omit) indicates that this zone file is for **xx.co.za**.

Therefore, all names you include in the zone file will be interpreted as **prefixes to xx.co.za**, and **must not** include xx.co.za. itself.

Leave any unused lines blank.

Question 4 [5 marks]

Assume the costs to transmit packets in a network are as follows (regardless of direction):

A–B: 1 C–D: 1

A–F: 10 D–E: 3

B–C: 6 E–F: 2

B–E: 1

You are applying **Dijkstra's algorithm** to determine the cheapest routes from **node A**.

- At the end of **iteration 1**, you determine that routing to **neighbour B** (at a cost of 1) is the cheapest alternative at this stage.
- At the start of **iteration 2**, you add **B** to the set of finalised nodes, and then consider the neighbours of this updated set.

Answer the following questions based on your continued application of Dijkstra's algorithm:

- a) Which node is picked as the cheapest alternative at the end of **iteration 3**?
- b) After completing iteration 3, but before starting iteration 4, what is the cost to reach **node C** from **node A**?
- c) After completing iteration 3, but before starting iteration 4, which node will (at this stage) finally deliver a packet en route to **node C**?
- d) Which node is added to the set of finalised nodes at the end of **iteration 4**?
- e) After completing iteration 4, but before starting iteration 5, what is the cost to reach **node C** from **node A**?

Question 5 [5 marks]

a) On Unix and Linux systems, **BIND** is a popular name server. What does the "d" in **BIND** stand for?

b) **Apache** is a well-known server — particularly, but not solely, on Linux systems. Apache is typically deployed to handle a specific protocol. Name the protocol. *(If the protocol has encrypted and non-encrypted variants, name the **variant that does not use encryption**.)*

c) What service does the **Telnet** protocol offer?

d) When a **DNS server** converts a domain name to an IP address, the correct terminology would be that the server _____ the domain name.

e) The Linux file /etc/hosts contains a list of pairs. Each pair consists of two entries separated by a space. The **second entry** is a name. What is the **first entry** of each pair?

Semester Test 2

Question 1

In each case, select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

1. If **MIME** is deemed to be a data communications protocol, it would fit best on layer ... of the ISO OSI model:

 - A. 7
 - B. 6
 - C. 5
 - D. 4
 - E. 3

2. Which character set is assumed to be understood by all parties involved in an HTTP exchange?

 - A. ASCII
 - B. EBCDIC
 - C. Unicode
 - D. UTF-8
 - E. ISO-8859-1

3. Consider the following protocol negotiation string in an HTTP request:
Accept-Charset: iso-8859-1, utf-8, utf-16, *;q=0.1
Suppose the server supports iso-8859-1, utf-8, and Shift_JIS. The response may then be encoded using:

 - A. iso-8859-1
 - B. utf-8
 - C. Shift_JIS
 - D. Any of the encodings listed as options above
 - E. One of two of the encodings listed as options above

4. Which of the following is **not** a valid MIME type?

 - A. binary
 - B. example
 - C. image
 - D. message
 - E. model

5. Suppose a message consists of JPG images embedded in HTML text. Which MIME type and subtype will be used to describe the message?
- A. text/plain
 - B. text/html
 - C. image/jpeg
 - D. multipart/mixed
 - E. multipart/alternative
6. Suppose an email consists of a message encoded in HTML and a JPG image (both included in the message). These two items are the only components of the email message. Which MIME type will be used to describe the entire email message?
- A. application/email
 - B. application/rfc822
 - C. multipart/alternative
 - D. multipart/mixed
 - E. text/html
7. How many Unicode characters can, in principle, be represented in UTF-8 using exactly four bytes? If the answer is expressed in the form 2^n , what is the value of n ?
- A. 14
 - B. 20
 - C. 21
 - D. 22
 - E. 27
8. Consider the byte sequence **D7 E5**. The Unicode character represented by this UTF-8 encoding is a(n):
- A. Accented character
 - B. Greek character
 - C. Emoji
 - D. Punctuation mark
 - E. D7 E5 is not a valid UTF-8 encoding
9. A BER value in ASN.1 is, in principle, encoded as a triple consisting of:
- A. A type, a subtype and a value
 - B. A type, a length and a value
 - C. A constructor and two operands
 - D. A variable name, as well as its minimum and maximum values
 - E. The same value encoded in binary, text, and hexadecimal

10. Which of the following is **not** a valid data type in ASN.1?
- A. STRUCT
 - B. CHOICE
 - C. SEQUENCE
 - D. ARRAY
 - E. INTEGER
11. The syntactic structure of an ASN.1 message is defined using:
- A. A grammar
 - B. An informal description in natural language
 - C. A diagrammatic depiction of the message
 - D. A bit pattern
 - E. A list of the types of the components of the message
12. The OSI layer that, among others, attempts to emulate database transactions where messages can be “rolled back” if the final message in a sequence is not sent, is layer:
- A. 2
 - B. 3
 - C. 4
 - D. 5
 - E. 6
13. Which of the following data streams generally work(s) better with an **unreliable** Layer 4 protocol?
- A. Web page
 - B. File transfer
 - C. Audio
 - D. More than one of the above
 - E. All of the above
14. Which of the following protocol(s) is/are used on the **transport layer** by **DNS**?
- A. TCP
 - B. UDP
 - C. IP
 - D. More than one of the above
 - E. All of the above
15. Which command enables one to see the status of **transport layer connections** on a host (in most operating systems)?
- A. netstat
 - B. ping
 - C. tracert / traceroute
 - D. tcp-show
 - E. ps

16. **UDP** uses the following flow-control mechanism:

- A. Stop and Wait
- B. Sliding Window
- C. Unrestricted
- D. More than one of the above
- E. All of the above

17. A sends a **TCP segment** to node B. The acknowledgement field contains the value **150**. The **ACK flag is not set**. This means:

- A. B may assume that bytes up to byte 149 that it had sent are acknowledged
- B. A should disregard the value 150 — it has no meaning
- C. This is a negative acknowledgement of the segment B has sent that had the sequence number 150
- D. The TCP layer at A is misconfigured; it should have set the ACK flag
- E. More than one of the above

18. Which of the following fields does **not** occur in a **TCP header**?

- A. Source port
- B. Window advertisement
- C. Sequence number
- D. Options
- E. Length

19. Which of the following statements is/are **false** about **phantom bytes** used in an initial 3-way TCP/IP handshake?

- A. It is counted as part of the SYN message
- B. It is counted as part of the SYN+ACK message
- C. It is counted as part of the ACK message
- D. It consists of one byte of data
- E. More than one of the statements above are false

20. A TCP connection in the **TIME-WAIT** state means that:

- A. The node is waiting for data from the node that it is connected to
- B. The connection is about to time-out if no further data is transmitted soon
- C. The connection will soon be established (as soon as the handshake is completed)
- D. The node is waiting for 'lost' traffic to arrive at the port that is no longer in use, before it will be available for reuse
- E. The network is congested

21. A TCP node that is in the **LISTEN** state:

- A. Is acting as a server and waiting for a SYN message
- B. Is acting as a client and has sent a SYN message
- C. May act as a client after sending a SYN message
- D. More than one of the above
- E. None of the above

22. What does the claim that **QUIC is a quick protocol** mean?
- A. It manages the lower layers to transmit raw data at higher bit rates
 - B. It reduces latency when establishing or re-establishing a connection
 - C. Where multiple parts of a message must be transported, it ensures that parts of the message are delivered quickly even if some parts are delayed
 - D. More than one of the above
 - E. All of the above
23. **QUIC connections** can migrate. Which party can initiate such a migration in **QUICv1** or **QUICv2**?
- A. The client
 - B. The server
 - C. The network management system
 - D. More than one of the above
 - E. All of the above
24. The **RIR for Europe** is:
- A. RIPE
 - B. EurNIC
 - C. Euronix
 - D. EurIN
 - E. EPIR
25. If one sends a message to an **IP multicast address**, the message will be delivered to:
- A. The node to which that IP address has been assigned
 - B. All nodes on the local network
 - C. All nodes that are members of some group
 - D. The network itself, rather than to a host on the network
 - E. The network gateway that connects the network to other networks

Question 2 [10 marks]

All values in this question are expressed in hexadecimal notation.

- Convert the following Unicode characters to byte sequences. Represent byte sequences as 2-digit hexadecimal numbers with a space between all bytes. (Your answers should resemble a format like: 12 bf 3a)

- a) U+1842
- b) U+10AD0
- c) U+2CA0

- UTF-8 encoding is self-synchronising. Consider the following byte sequence, which is not a valid UTF-8 encoding:

47	CE	B5	C4	85	E1	80	AA	E1	9C	A0
BA	AD	F0	90	8F	8C	57	73	E1	83	A8

- d) Provide the first valid Unicode character that occurs in this sequence. (Format: U+xxxx)
 - e) Provide the second valid Unicode character that occurs in this sequence. (Format: U+xxxx)
 - f) Provide the subsequence of bytes that are invalid.
 - g) Convert the bytes starting at the point where synchronisation is re-established, extract the first Unicode character formed (Format: U+xxxx)
 - h) Provide the last valid Unicode character in this sequence. (Format: U+xxxx)
- We did not discuss UTF-16 and UTF-32 in detail, but UTF-16 represents Unicode characters using 16-bit words. UTF-32 uses 32-bit words. When represented as bytes, endianness matters: Big-endian means 1234 → 12 34, and 12345678 → 12 34 56 78. Use big-endian representations for the following:
- i) Represent U+0048 in UTF-16 as a byte sequence
 - ii) Represent U+0F40 in UTF-32 as a byte sequence

Question 3 [5 marks]

Very briefly state what happens when the following **TCP timers** expire.

Keep your answers short and precise. For example, if expiration of a timer causes a segment to be transmitted, simply state: *A segment containing data is transmitted.*

- a) Retransmission timer
- b) Acknowledgement timer
- c) Persistence timer
- d) Keepalive timer
- e) Quiet timer

Question 4 [5 marks]

TCP is a reliable protocol. List five mechanisms used by TCP to provide this reliability. Keep your answers brief; no explanation are required.

Question 5 [5 marks]

Answer the following questions about IPv4 addresses. Use CIDR notation for all your answers.

- a) Which IPv4 address is used in a routing table to indicate that the routing table entry points to the default gateway?
- b) A block of 256 'class C' network addresses is reserved for private use. Provide the first-class C network address that occurs within this block.
- c) A block of 16 'class B' network addresses is reserved for private use. Provide the last class B network address that occurs within this block.
- d) 137.215.98.140 is a host on the University of Pretoria network. What is the UP-network address?
- e) 8.8.8.8 is a popular DNS server provided by Google. If this address were still part of a class-based address, what would the broadcast address on this network be?

Please recall that CIDR notation has to be used for all your answers in this question.